Master Business Analytics

2025 - 2026

Professional vs dual variant

Core courses Business Analytics (compulsory)	66 EC							
Course Name	*Code*	*EC*	*P1*	*P2*	*P3*	*P4*	*P5*	*P6*
Master Project BA	X_400459	36						
Research Seminar BA	XM_0054	6				3	3	
Advanced Machine Learning	XM_0010	6	6					
Applied Stochastic Modeling	X_400392	6	3	3				
Statistical Models	X_400418	6	3	3				
Compulsory Professional variant (elective for Dual variant)								
Project OBP	X_400213	6			6			
Compulsory Dual variant								
Dual Work period	XM_41010	12						

Specialization: Computational Intelligence

Compulsory courses	18 EC							
Course Name	*Code*	*EC*	*P1*	*P2*	*P3*	*P4*	*P5*	*P6*
Evolutionary Computing	X_400111	6	6					
Natural Language Processing Technology	XM_0121	6					6	
Deep Learning	XM_0083	6		6				
Total			18	12	6	3	9	0

Compulsory selection	18 EC (12 EC for dual or professional with semester abroad)							
Course Name	*Code*	*EC*	*P1*	*P2*	*P3*	*P4*	*P5*	*P6*
Performance of Networked Systems	X_405105	6		6				
Data Mining Techniques	X_400108	6					6	
ML for the quantified self	XM_40012	6						6
Multi-agent Systems	XM_0052	6		6				
Dynamic Programming and Reinforcement Learning	XM_0093	6		6				
Learning Machines*	XM_0061	6			6			6

offered in p3 and p6, limited capacity

Specialization: Optimization of Business Processes

Compulsory courses	18 EC							
Course Name	*Code*	*EC*	*P1*	*P2*	*P3*	*P4*	*P5*	*P6*
Mathematical Optimization	XM_0051	6		6				
Optimization of Business Processes	X_400422	6				3	3	
DPRL	XM_0093	6		6				
			0	12	0	3	3	0

Compulsory selection	18 EC (12 EC for dual or professional with semester abroad)							
Course Name	*Code*	*EC*	*P1*	*P2*	*P3*	*P4*	*P5*	*P6*
Continuous Optimization (LNMB)	XMM_400446	6	3	3				
Discrete Opt (LNMB)	XMM_400445		3	3				
Scheduling (LNMB)	XMM_400396	6				3	3	
Advanced Linear Programming (LNMB)	XMM_400326	6				3	3	
Evolutionary Computing	X_400111	6	6					
Performance of Networked Systems	X_405105	6		6				
Data Mining Techniques	X_400108	6					6	
Entrepreneurship in Analytics and AI		6				6		
Applied Analysis: Financial Mathematics (MSc)	XM_0140	6				3	3	
Supply Chain Lab		6				6		

Specialization: Financial Risk Management

Compulsory courses	18 EC							
Course Name	*Code*	*EC*	*P1*	*P2*	*P3*	*P4*	*P5*	*P6*
Stochastic Processes for Finance	X_400352	6	3	3				
Applied Analysis: Financial Mathematics (MSc)	XM_0140	6				3	3	
Investments	E_EBE3_INV	6					6	
			3	3	0	3	9	0

Compulsory selection	18 EC (12 EC for dual or professional with semester abroad)							
Course Name	*Code*	*EC*	*P1*	*P2*	*P3*	*P4*	*P5*	*P6*
Advanced Asset Pricing**	E_FIN_AAP	6		6				
Quantitative Investing**	E_FIN_QI	6	6					
Financial Markets and Institutions*	E_FIN_FMI	6		6				
Derivatives*	E_FIN_DER	6				6		
Quantitative Financial Risk Management*	E_FIN_QFRM	6					6	
Mathematical Optimization	XM_0051	6		6				
DPRL	XM_0093	6		6				
Data Mining Techniques	X_400213	6					6	

**course Investments is a compulsory prerequisite*

*** out of Advanced Asset Pricing and Quantititave Investing, only one is allowed*

Electives

Electives	18 EC							
Course Name	*Code*	*EC*	*P1*	*P2*	*P3*	*P4*	*P5*	*P6*
Programming Large-Scale Parallel Systems	XM_40017	6	6					
Data Mining Techniques	X_400108						6	
Applied Text Mining 1: Methods	L_PAMATLW004				6			
Natural Language Processing Technology	XM_0121	6					6	
Deep Learning	XM_0083	6		6				
ML for the quantified self	XM_40012	6						6
Learning Machines***	XM_0061	6			6			6
Continuous Optimization (LNMB)	XMM_400446	6	3	3				
Discrete Optimization (LNMB)	XMM_400445		3	3				
Scheduling (LNMB)	XMM_400396	6				3	3	
Advanced Linear Programming (LNMB)	XMM_400326	6				3	3	
Queueing Theory	XMM_400397					3	3	
Performance of Networked Systems	X_405105			6				
Mathematical Optimization	XM_0051	6		6				
Optimization of Business Processes	X_400422	6				3	3	
Project OBP	X_400213	6			6			
Enterpreneurship in Analytics and AI	XM_0090	6				6		
Supply Chain Lab	E_TSCM_SCL	6				6		
Stochastic Processes for Finance	X_400352	6	3	3				
Applied Analysis: Financial Mathematics (MSc)	XM_0140	6				3	3	
Investments	E_EBE3_INV	6					6	
Advanced Asset Pricing**	E_FIN_AAP	6		6				
Quantitative Investing**	E_FIN_QI	6	6					
Financial Markets and Institutions*	E_FIN_FMI	6		6				
Time Series Models	E_EORM_TSM	6				6		
Project Reinforcement Learning	XM_0120				6			

**course Investments is a compulsory prerequisite*

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****offered in p3 and p6, limited capacity*

Recommendations on the order of courses:

- If you wish to follow Mathematical Optimization and Supply Chain Lab, it is optimal to do Supply Chain Lab first and Mathematical Optimization next.

- If you wish to follow Mathematical Optimization and Discrete/Continuous Optimization, it is optimal to do Mathematical Optimization first and Discrete/Continuous Optimization next.

- If you wish to follow Project Optimization of Business Processes and the course Optimization of Business Processes, it is optimal to follow the project first and the course next.

- If you wish to follow Project Reinforcement Learning, it is optimal to do Dynamic Programming and Reinforcement Learning first.

- It is optimal to follow Deep Learning in year 2.